

UMUTCAN KARAKAŞ

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Education

Istanbul Technical University

Ph.D. in Computer Engineering
M.Sc. in Computer Engineering
B.Sc. in Computer Engineering

Feb. 2024 – Present
Sep. 2021 – Feb. 2024
Sep. 2015 – Aug. 2020

Anadolu University

A.A. in Web Design and Coding

Sep. 2016 – Aug. 2018

Experience

DefineX Consulting | Technology | Labs

Senior AI Engineer

Sep. 2025 – Present

Senior .NET Developer

Apr. 2024 – Dec. 2025

- As Senior AI Engineer, developed document intelligence pipelines for İş Bankası, converting independent audit report PDFs into structured, machine-readable data using OCR and LLM-based processing.
- Currently serving as Project Owner on DefineX's side for the Garanti Bankası + Mastercard collaboration, leading the design and delivery of an AI-agent-based system that assists users with clothing, travel, and marketplace purchases directly within the banking application.
- Previously contributed as Senior .NET Developer to large-scale call center modernization projects for İş Bankası and Getir Finans, delivering end-to-end backend solutions, third-party integrations, and production-ready features.

Kaizen Technology

Dec. 2022 - Apr. 2023

.NET Developer

- Developed key backend features for the Anadolu Efes Basketball Team mobile app, including real-time score tracking, player/team statistics, loyalty points, and fan reward systems.
- Maintained the Nutella app, enabling prize redemption through customer-earned points.

Erstream Video Delivery Company

Jan. 2020 – Nov. 2022

Software Developer

- Built internal monitoring and notification services to improve system reliability and issue tracking.
- Led the development of a user management and analytics panel for Exxen, focusing on data visualization, role-based permissions, and security.

Rhino Runner

March 2019 - June 2019

Junior .NET Developer

Sociable, Inc.

Summer 2018

Software Developer

Ayedas, EnerjiSA

Summer 2017

Business Logic Intern

Projects

Automated Fairness Testing with Representative Sampling

- Developed RSFair, a machine learning fairness tool using OMP and K-SVD for advanced representative sampling.
- Enhanced fairness assessment accuracy and efficiency, outperforming traditional methods.
- Achieved significant improvements in bias detection and fairness enhancement in ML models.
- Can be viewed from <https://doi.org/10.1145/3617555.3617871>

Skills

Backend & Platform Engineering: .NET, RESTful APIs, distributed services, system integration, enterprise systems

Web Development: JavaScript, TypeScript, HTML/CSS, Kendo UI, jQuery

Applied AI & LLM Systems: OCR-based document processing pipelines, LLM-assisted text extraction and structuring, AI-powered in-app assistants

Data Stores & Search: MySQL, Elasticsearch